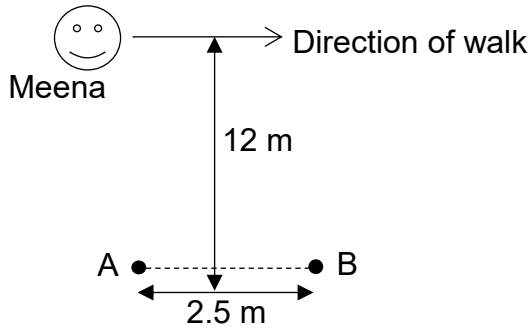
3 Two identical sound speakers placed at positions A and B are connected to the same signal generator, 2.5 m apart. Meena walks in a direction parallel to AB, 12 m away, as shown in the figure below:



As she walks, she detects a series of loud and soft sounds.

(a) Adjacent loud regions are 0.60 m apart. Calculate the frequency of the sound used. The speed of sound is 340 m s^{-1}. [3]

(b) At one of the loud regions, Meena detects an intensity of *I*. The amplitude of each wave from each speaker is *A*.

(i) State the resultant wave amplitude at Meena's location. [1]

The power output of one of the speakers is reduced to half its original value, while the other speaker remains unchanged.

(ii) For the speaker with the reduced power output, state the amplitude of its wave at Meena's location in terms of *A*. [1]

(iii) Calculate the new resultant wave amplitude at Meena's location in terms of *A*. [1]

(iv) Calculate the new intensity detected by Meena in terms of I. [2]